

Student Feedback Satisfaction Analysis

Introduction

This notebook analyzes student feedback data to understand overall satisfaction levels. The analysis is based on ratings provided on a 1–5 scale. The objective is to clean the data, classify satisfaction levels, derive sentiment, and summarize key insights.

```
from google.colab import files
uploaded = files.upload()

<IPython.core.display.HTML object>

Saving student_feedback.csv to student_feedback.csv

import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('student_feedback.csv')

df.head()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 1001,\n  \"fields\": [\n    {\n      \"column\": \"Unnamed: 0\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 289,\n        \"min\": 0,\n        \"max\": 1000,\n        \"num_unique_values\": 1001,\n        \"samples\": [\n          521,\n          941,\n          741\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Student ID\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 289,\n        \"min\": 0,\n        \"max\": 1000,\n        \"num_unique_values\": 1001,\n        \"samples\": [\n          765,\n          677,\n          537\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Well versed with the subject\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1,\n        \"min\": 5,\n        \"max\": 10,\n        \"num_unique_values\": 6,\n        \"samples\": [\n          5,\n          6,\n          10\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Explains concepts in an understandable way\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2,\n        \"min\": 2,\n        \"max\": 10,\n        \"num_unique_values\": 9,\n        \"samples\": [\n          9,\n          5,\n          3\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Use of presentations\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
```



```

null    int64
4    Use of presentations    1001 non-
null    int64
5    Degree of difficulty of assignments    1001 non-
null    int64
6    Solves doubts willingly    1001 non-
null    int64
7    Structuring of the course    1001 non-
null    int64
8    Provides support for students going above and beyond    1001 non-
null    int64
9    Course recommendation based on relevance    1001 non-
null    int64
dtypes: int64(10)
memory usage: 78.3 KB

```

```
df.describe(include='all')
```

```

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"Unnamed: 0\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 360.546420471925,\n        \"min\": 0.0,\n        \"max\": 1001.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          1001.0,\n          500.0,\n          750.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Student ID\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 360.546420471925,\n        \"min\": 0.0,\n        \"max\": 1001.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          1001.0,\n          500.0,\n          750.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Well versed with the subject\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 351.5330025154177,\n        \"min\": 1.6929984513412213,\n        \"max\": 1001.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          7.497502497502498,\n          8.0,\n          1001.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Explains concepts in an understandable way\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 351.9635045408746,\n        \"min\": 2.0,\n        \"max\": 1001.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          6.081918081918082,\n          6.0,\n          1001.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Use of presentations\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 352.0257536891051,\n        \"min\": 1.4158528739383012,\n        \"max\": 1001.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          5.942057942057942,\n          6.0,\n          1001.0\n        ],\n        \"semantic_type\": \"\",

```

```

{"description": "\n    },\n    {\n    \"column\": \"Degree of difficulty of assignments\", \"properties\": {\n    \"dtype\": \"number\", \"std\": 352.13595404294387, \"min\": 1.0, \"max\": 1001.0, \"num_unique_values\": 8, \"samples\": [\n    5.430569430569431, 5.0, 1001.0\n    ], \"semantic_type\": \"\", \"description\": \"\n    },\n    {\n    \"column\": \"Solves doubts willingly\", \"properties\": {\n    \"dtype\": \"number\", \"std\": 352.0831011075789, \"min\": 1.0, \"max\": 1001.0, \"num_unique_values\": 8, \"samples\": [\n    5.474525474525475, 6.0, 1001.0\n    ], \"semantic_type\": \"\", \"description\": \"\n    },\n    {\n    \"column\": \"Structuring of the course\", \"properties\": {\n    \"dtype\": \"number\", \"std\": 352.0726062200091, \"min\": 1.0, \"max\": 1001.0, \"num_unique_values\": 8, \"samples\": [\n    5.6363636363637, 6.0, 1001.0\n    ], \"semantic_type\": \"\", \"description\": \"\n    },\n    {\n    \"column\": \"Provides support for students going above and beyond\", \"properties\": {\n    \"dtype\": \"number\", \"std\": 352.07276620844857, \"min\": 1.0, \"max\": 1001.0, \"num_unique_values\": 8, \"samples\": [\n    5.662337662337662, 6.0, 1001.0\n    ], \"semantic_type\": \"\", \"description\": \"\n    },\n    {\n    \"column\": \"Course recommendation based on relevance\", \"properties\": {\n    \"dtype\": \"number\", \"std\": 352.07624563694816, \"min\": 1.0, \"max\": 1001.0, \"num_unique_values\": 8, \"samples\": [\n    5.5984015984015985, 6.0, 1001.0\n    ], \"semantic_type\": \"\", \"description\": \"\n    }\n    ], \"type\": \"dataframe\"}

```

```
df.isnull().sum()
```

Unnamed: 0	0
Student ID	0
Well versed with the subject	0
Explains concepts in an understandable way	0
Use of presentations	0
Degree of difficulty of assignments	0
Solves doubts willingly	0
Structuring of the course	0
Provides support for students going above and beyond	0
Course recommendation based on relevance	0
dtype: int64	

```
df = df.dropna(subset=['Overall_satisfaction'])
```

```
df = df.reset_index(drop=True)
print(df.head())
```

	Unnamed: 0	Student ID	Well versed with the subject \
0	0	340	5
1	1	253	6
2	2	680	7
3	3	806	9
4	4	632	8

	Explains concepts in an understandable way	Use of presentations \
0	2	7
1	5	8
2	7	6
3	6	7
4	10	8

	Degree of difficulty of assignments	Solves doubts willingly \
0	6	9
1	6	2
2	5	4
3	1	5
4	4	6

	Structuring of the course \
0	2
1	1
2	2
3	9
4	6

	Provides support for students going above and beyond \
0	1
1	2
2	3
3	4
4	9

	Course recommendation based on relevance
0	8
1	9
2	1
3	6
4	9

```
type(df)
```

```
pandas.core.frame.DataFrame
```

```
df.columns
```

```
Index(['Unnamed: 0', 'Student ID', 'Well versed with the subject',
      'Explains concepts in an understandable way', 'Use of
presentations',
      'Degree of difficulty of assignments', 'Solves doubts
willingly',
      'Structuring of the course',
      'Provides support for students going above and beyond',
      'Course recommendation based on relevance'],
      dtype='object')
```

```
len(df.columns)
```

```
10
```

```
df.columns = [col.strip().lower().replace(' ', '_') for col in
df.columns]
```

```
print(df.columns.tolist())
```

```
['unnamed_0', 'student_id', 'well_versed_with_the_subject',
'explains_concepts_in_an_understandable_way', 'use_of_presentations',
'degree_of_difficulty_of_assignments', 'solves_doubts_willingly',
'structuring_of_the_course',
'provides_support_for_students_going_above_and_beyond',
'course_recommendation_based_on_relevance']
```

```
rating_columns = [
    'course_recommendation_based_on_relevance',
    'explains_concepts_in_an_understandable_way',
    'provides_support_for_students_going_above_and_beyond',
    'well_versed_with_the_subject'
]
```

```
print(df.columns.tolist())
```

```
['unnamed_0', 'student_id', 'well_versed_with_the_subject',
'explains_concepts_in_an_understandable_way', 'use_of_presentations',
'degree_of_difficulty_of_assignments', 'solves_doubts_willingly',
'structuring_of_the_course',
'provides_support_for_students_going_above_and_beyond',
'course_recommendation_based_on_relevance']
```

```
df['Overall_satisfaction'] = df[rating_columns].mean(axis=1)
```

```
print(df[['student_id', 'Overall_satisfaction'][:10])
```

	student_id	Overall_satisfaction
0	340	4.00
1	253	5.50
2	680	4.50
3	806	6.25
4	632	9.00

5	832	3.50
6	772	5.25
7	961	7.50
8	814	6.75
9	863	5.75

```
def satisfaction_level(score):
    if score <= 2:
        return "Low"
    elif 3 <= score <= 4:
        return "Medium"
    else: # score = 5
        return "High"
```

```
df['Satisfaction_Level'] =
df['Overall_satisfaction'].apply(satisfaction_level)
```

```
print(df[['student_id', 'Overall_satisfaction',
'Satisfaction_Level']].head(11))
```

	student_id	Overall_satisfaction	Satisfaction_Level
0	340	4.00	Medium
1	253	5.50	High
2	680	4.50	High
3	806	6.25	High
4	632	9.00	High
5	832	3.50	Medium
6	772	5.25	High
7	961	7.50	High
8	814	6.75	High
9	863	5.75	High
10	576	7.00	High

```
def sentiment(level):
    if level == "High":
        return "Positive"
    elif level == "Medium":
        return "Neutral"
    else: # Low
        return "Negative"
```

```
df['Sentiment'] = df['Satisfaction_Level'].apply(sentiment)
```

```
# Check result
```

```
print(df[['student_id', 'Overall_satisfaction', 'Satisfaction_Level',
'Sentiment']].head(11))
```

	student_id	Overall_satisfaction	Satisfaction_Level	Sentiment
0	340	4.00	Medium	Neutral
1	253	5.50	High	Positive
2	680	4.50	High	Positive

3	806	6.25	High	Positive
4	632	9.00	High	Positive
5	832	3.50	Medium	Neutral
6	772	5.25	High	Positive
7	961	7.50	High	Positive
8	814	6.75	High	Positive
9	863	5.75	High	Positive
10	576	7.00	High	Positive

```
print(df['Sentiment'].value_counts())
```

```
Sentiment
Positive    950
Neutral      51
Name: count, dtype: int64
```

```
df['Satisfaction_Level'].value_counts()
```

```
Satisfaction_Level
High    950
Medium   51
Name: count, dtype: int64
```

```
total_responses = len(df)
```

```
satisfaction_summary = df['Satisfaction_Level'].value_counts()
sentiment_summary = df['Sentiment'].value_counts()
```

```
print("Total Responses:", total_responses)
print("\nSatisfaction Level Summary:\n", satisfaction_summary)
print("\nSentiment Summary:\n", sentiment_summary)
```

```
Total Responses: 1001
```

```
Satisfaction Level Summary:
Satisfaction_Level
High    950
Medium   51
Name: count, dtype: int64
```

```
Sentiment Summary:
Sentiment
Positive    950
Neutral      51
Name: count, dtype: int64
```

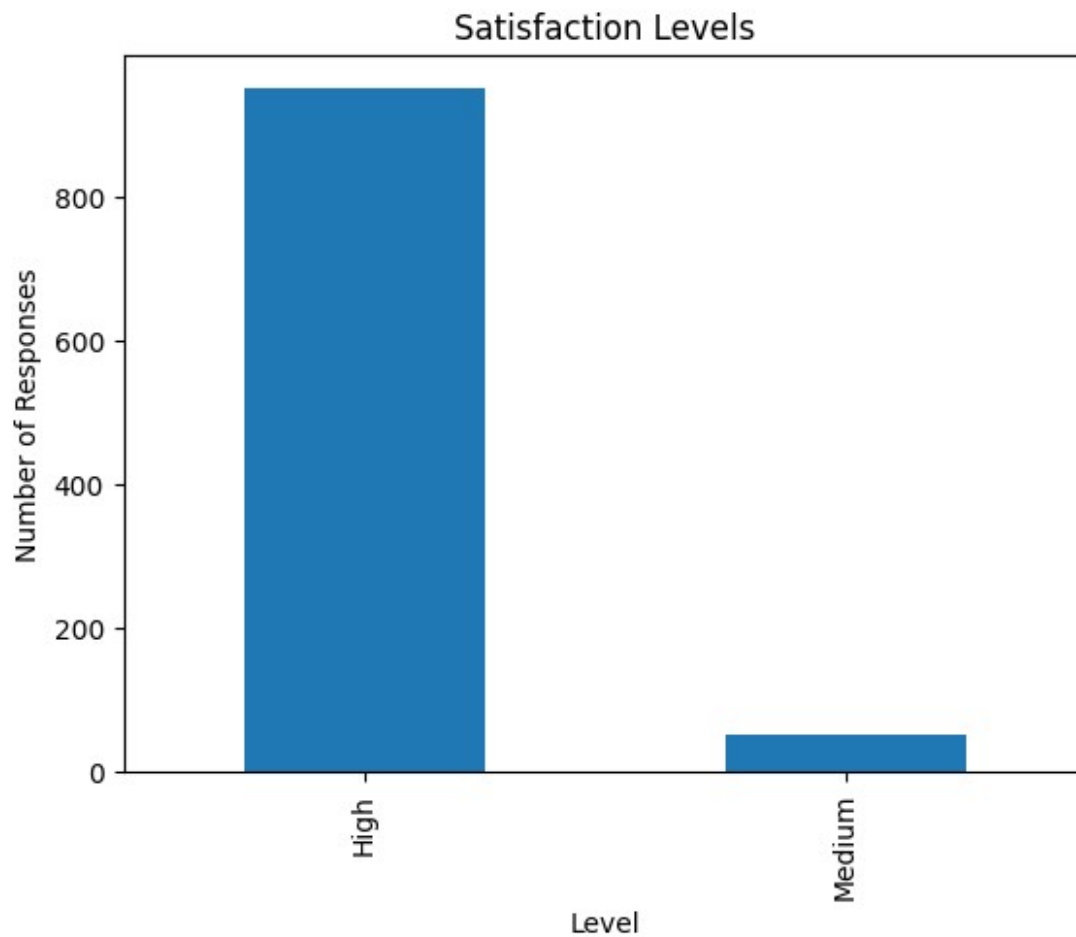
```
import matplotlib.pyplot as plt
```

```
satisfaction_summary.plot(kind='bar')
plt.title('Satisfaction Levels')
plt.xlabel('Level')
```

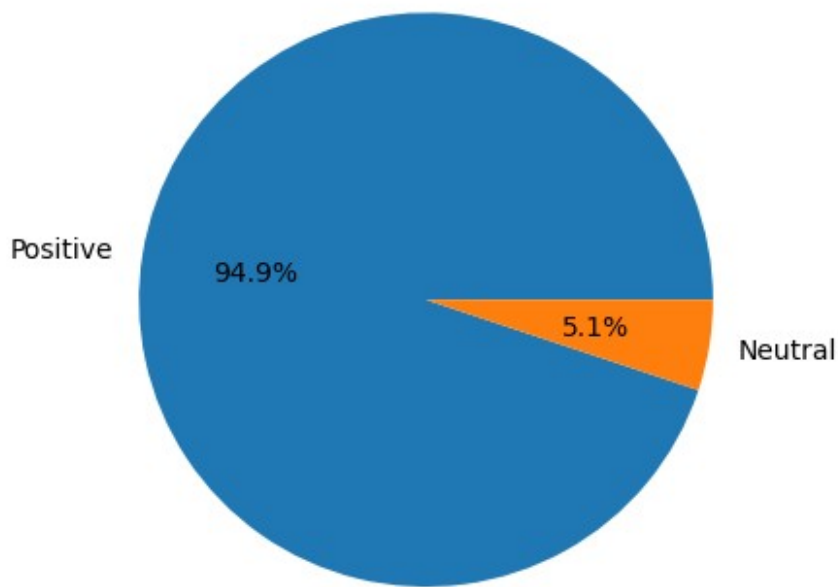


```
plt.ylabel('Number of Responses')
plt.show()

sentiment_summary.plot(kind='pie', autopct='%1.1f%%')
plt.title('Sentiment Distribution')
plt.ylabel('')
plt.show()
```



Sentiment Distribution



Summary

The dataset was cleaned by removing duplicate records and missing values. Student satisfaction ratings were analyzed using a 1–5 scale. Ratings were classified into Low, Medium, and High satisfaction levels. The majority of students reported high satisfaction, indicating a positive overall response. Only a small proportion of students expressed low satisfaction.

Conclusion

The analysis indicates that students are generally satisfied based on their ratings. High satisfaction levels dominate the responses, suggesting effective institutional practices. Future improvements can focus on addressing the concerns of moderately and less satisfied students.